

Nobel Dhar

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Research Interests

Systems for efficient machine learning: large language model (LLM) inference on memory-constrained hardware, storage-aware inference, activation sparsity and sparsity prediction, and memory management for mixture-of-experts (MoE) models.

Education

Kennesaw State University Aug 2023 – May 2027 (expected)
Ph.D., Computer Science Marietta, GA

- Dissertation: *Bridging the Model–Memory Gap: Intelligent Storage I/O for Scalable Neural Network Inference on Resource-Constrained Platforms*
- Advisor: Dr. Kun Suo · Co-advisor: Dr. Bobin Deng · Dissertation proposal defended May 2026

Shahjalal University of Science and Technology 2016 – 2021
B.Sc., Computer Science and Engineering Sylhet, Bangladesh

Research Experience

Graduate Research Assistant Aug 2023 – Present
Department of Computer Science, Kennesaw State University Marietta, GA

- **Edge LLM characterization** [C7]. Deployed Llama-2-7B (INT4) on edge platforms from the Raspberry Pi 4B to the NVIDIA Jetson AGX Orin, and identified insufficient memory and computing resources as the main obstacles to on-device LLM inference.
- **Activation sparsity** [C5]. Created up to 50% feed-forward activation sparsity without retraining in LLMs that have little natural sparsity (Llama-3-8B, Mistral-7B, and Phi-3), with negligible accuracy loss at 30% sparsity and acceptable perplexity at 50%.
- **Sparsity prediction** [C2]. Reformulated per-neuron activation prediction as classification over a bank of activation-pattern centroids, which makes sparsity prediction practical at LLM scale.
- **Storage-aware LLM inference** [C1]. Designed and built NeuroPrefetcher, a runtime that runs LLMs larger than the available device memory by predicting the feed-forward neurons each token needs and reading only the newly required weight rows from NVMe storage (`io_uring`, `O_DIRECT`) into GPU buffers. Trained its sparsity predictor from scratch with distributed data-parallel training (PyTorch DDP/NCCL) on 8 NVIDIA A100 GPUs. In a capability study of 20 production and research inference systems on an NVIDIA Jetson AGX Orin under a 14 GiB memory cap, only four generated output, and NeuroPrefetcher was the fastest; across constrained memory budgets it runs 7.9–12.0× faster than llama.cpp.
- **Current work.** Memory management for mixture-of-experts inference on edge platforms (manuscript in preparation).

Publications

Citation metrics ([Google Scholar](#), October 2026): 168 citations, h-index 6.

Peer-Reviewed Conference Papers

- [C1] **N. Dhar**, M. R. Islam, X. Zhang, G. Sun, S. Islam, B. Deng, and K. Suo. “NeuroPrefetcher: Storage-Aware Sparse LLM Inference via Delta Prefetching.” In *Proceedings of the 55th International Conference on Parallel Processing (ICPP '26)*, ACM, 2026. Acceptance rate 23% (117 of 509). [\[DOI\]](#) [\[arXiv\]](#) [\[Code\]](#)
- [C2] **N. Dhar**, B. Deng, M. R. Islam, X. Zhang, K. F. A. Nasif, and K. Suo. “A Sparsity Predicting Approach for Large Language Models via Activation Pattern Clustering.” In *Euro-Par 2025: Parallel Processing*, Lecture Notes in Computer Science, vol. 15900, Springer, 2025. [\[DOI\]](#) [\[Code\]](#)
- [C3] K. F. A. Nasif, B. Deng, L. Chen, Y. Xie, S. Teng, J. Li, L. Zhao, S. M. S. Alam, **N. Dhar**, K. Suo, and D. C. T. Lo. “AccelMD: A Self-Adaptive AI-Enabled Framework for Accelerating Molecular Dynamics Simulations.” In *Proceedings of the IEEE International Conference on Tools with Artificial Intelligence (ICTAI '25)*, 2025. [\[DOI\]](#)
- [C4] M. R. Islam, B. Deng, **N. Dhar**, T. N. Nguyen, S. He, Y. Shi, and K. Suo. “Characterizing and Understanding Energy Footprint and Efficiency of Small Language Model on Edges.” In *Proceedings of the IEEE International Conference on Mobile Ad-Hoc and Smart Systems (MASS '25)*, 2025. [\[DOI\]](#)
- [C5] **N. Dhar**, B. Deng, M. R. Islam, K. F. A. Nasif, L. Zhao, and K. Suo. “Activation Sparsity Opportunities for Compressing General Large Language Models.” In *Proceedings of the IEEE International Performance, Computing, and Communications Conference (IPCCC '24)*, 2024. [\[DOI\]](#) [\[Code\]](#)
- [C6] M. R. Islam, **N. Dhar**, B. Deng, T. N. Nguyen, S. He, and K. Suo. “Characterizing and Understanding the Performance of Small Language Models on Edge Devices.” In *Proceedings of the IEEE International Performance, Computing, and Communications Conference (IPCCC '24)*, 2024. [\[DOI\]](#)
- [C7] **N. Dhar**, B. Deng, D. Lo, X. Wu, L. Zhao, and K. Suo. “An Empirical Analysis and Resource Footprint Study of Deploying Large Language Models on Edge Devices.” In *Proceedings of the ACM Southeast Conference (ACMSE '24)*, 2024. [\[DOI\]](#) [\[Data & Code\]](#)
- [C8] M. R. Islam, L. Vu, **N. Dhar**, B. Deng, and K. Suo. “Building a Resilient and Sustainable Grid: A Study of Challenges and Opportunities in AI for Smart Virtual Power Plants.” In *Proceedings of the ACM Southeast Conference (ACMSE '24)*, 2024. [\[DOI\]](#)
- [C9] K. Suo, L. Vu, M. R. Islam, **N. Dhar**, T. N. Nguyen, S. He, and X. Wu. “A Systematic Investigation of Hardware and Software in Electric Vehicular Platform.” In *Proceedings of the ACM Southeast Conference (ACMSE '24)*, 2024. [\[DOI\]](#)
- [C10] L. Vu, K. Suo, M. R. Islam, **N. Dhar**, T. N. Nguyen, S. He, and Y. Shi. “Living on the Electric Vehicle and Cloud Era: A Study of Cyber Vulnerabilities, Potential Impacts, and Possible Strategies.” In *Proceedings of the ACM Southeast Conference (ACMSE '24)*, 2024. [\[DOI\]](#)
- [C11] **N. Dhar**, G. Saha, P. Bhattacharjee, A. Mallick, and M. S. Islam. “Pointer over Attention: An Improved Bangla Text Summarization Approach Using Hybrid Pointer Generator Network.” In *Proceedings of the International Conference on Computer and Information Technology (ICCIT '21)*, 2021. [\[DOI\]](#)

Conference Presentations

“NeuroPrefetcher: Storage-Aware Sparse LLM Inference via Delta Prefetching.” ICPP 2026, Singapore	Sep 2026
“A Sparsity Predicting Approach for Large Language Models via Activation Pattern Clustering.” Euro-Par 2025, Dresden, Germany	Aug 2025
“Activation Sparsity Opportunities for Compressing General Large Language Models.” IEEE IPCCC 2024, Orlando, FL	Nov 2024
“An Empirical Analysis and Resource Footprint Study of Deploying Large Language Models on Edge Devices.” ACMSE 2024, Marietta, GA	Apr 2024
“Pointer over Attention: An Improved Bangla Text Summarization Approach Using Hybrid Pointer Generator Network.” ICCIT 2021, Dhaka, Bangladesh	Dec 2021

Honors and Awards

Graduate student travel grants (internal), Kennesaw State University

Teaching Experience

Grading assistant for Dr. Kun Suo’s courses, Department of Computer Science, Kennesaw State University

Professional Service

Reviewer, IEEE International Conference on Tools with Artificial Intelligence (ICTAI) 2025
Sub-reviewer, assisting Dr. Kun Suo with paper reviews for more than 10 conferences

Industry Experience

Software Engineer Aug 2021 – Jun 2023
Bangladesh Japan Information Technology (BJIT) Dhaka, Bangladesh

- Android Open Source Project (AOSP) system software, down to the hardware abstraction layer, and the cloud-backed translation app for PockeTalk, a commercial handheld translation device.

Associate Specialist, Software Development Feb 2021 – Aug 2021
ParallaxLogic Infotech Dhaka, Bangladesh

- Android applications (Kotlin) for ekShop, a Bangladesh government e-commerce platform.

Technical Skills

Programming: Python, C++, CUDA C++, Java, Kotlin

Tools and platforms: PyTorch (DDP, NCCL), Linux storage I/O (io_uring, O_DIRECT, NVMe), mmap/madvise, cgroups, Nsight, NVIDIA Jetson

References

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Dr. Xuechen Zhang
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Collaborator; co-author of [C1]